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ALP-CRS: enhancing course recommendation in online education through data mining and graph analysis

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ABSTRACT

Recommender systems (RSs) are an increasingly common type of intelligent software that predicts the preferences of users. Online learning has become a ubiquitous and necessary means of lifelong learning, which provides flexibility in time and place. This paper introduces a course recommendation system (ALP-CRS) which is founded on APRIORI and Link Prediction. It employs link prediction and graph clustering techniques to enhance the accuracy of the recommendations, and employs association rule mining to detect common trends in the course selection. Three stages make up the methodology: outlining the guidelines for choosing courses, comparing student characteristics, and making suggestions. In the first stage, features are extracted from the course selection data and course selection rules are derived. In order to improve recommendations using link prediction, the next step comprises creating student graphs, grouping them, and filtering comparable students. During the last stage, the system offers recommendations on the filtered information, which gives students a more targeted and personalized experience of learning. According to the findings, the suggested approach was capable of providing suggestions with an 82% precision rate and a 24% recall rate, meaning that it outperformed the compared approaches by at least 4% in precision and 10% in recall.

1. Introduction

In the past, printed textbooks were the main instructional materials. But in the contemporary world, the spread of computers, the World Wide Web and the explosion of information has led to the need to create systems that offer intelligent suggestions, thus making the decision-making processes easier. Accurate linguistic description in predictive modeling plays an important role in improving the understanding of these technical contributions [1]. Therefore, the recommender systems field has developed considerably over the past 2 decades [2]. This technological change has easily penetrated into the education sector, leading to the sudden development of online learning platforms. especially due to the COVID-19, online learning has grown rapidly across the globe. Over 10 million people utilize Coursera, edX, XuetangX, and other online education sites on a regular basis. The rapid expansion of online learning has altered traditional teaching methods and made it possible for students to learn at any time, from any location.

Online learning has grown in importance as a model for education and as a technological tool for students to advance their academic

education and gain new skills [3]. This is because there is a plethora of learning tools available to online learners, who are not constrained by time or location. These systems must be framed contextually to comprehend their motivation as recent research on student rating systems has pointed out [4]. Despite the fact that a lot of study has been done on the subject of recommended online learning tools, there are still certain restrictions. Resources, for instance, are abundant, but there are a lot of variances in their quality and level, making it difficult for learners to determine whether a resource is suited for them or has been mastered.

Nevertheless, the abundance of resources poses a major challenge. It is a very challenging task to develop a proper knowledge framework because most learning materials are nonlinear, which contributes to information overload [5]. According to recent studies on classification models [6], structured transition of narratives is critical in rationale methodological decisions in solving such problems. Thus, Recommender Systems (RSs) have become essential in the technology-enhanced learning to counter this overload. They are used to identify and suggest relevant learning objects to the student—solving problems where,

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for example, a learning object spans multiple categories or disciplines [7]. Before recommending information that sparks the user's attention, a customized learning system might analyze user data to ascertain the user's selected knowledge.

In the information era, personalized learning is an unavoidable trend. Machine learning encompasses the creation, examination, and research of algorithms that can automatically identify patterns in data and utilize that information for predicting future data or make decisions. These online education systems continually modify and enhance their recommendations by utilizing machine learning algorithms. This allows them to provide students a carefully chosen range of courses that are tailored to their individual requirements and learning objectives.

1.1. Problem Statement and Motivation

Despite the tremendous advancement in the online learning environment and the application of the recommender systems, several limitations still exist which make the application of more advanced and sophisticated methods of recommendation justifiable. Although abundance of resources is a well-known advantage, the heterogeneity of the resources tends to create a serious problem of aligning the content with the needs of particular learners, which has been formulated in recent predictive analytics studies [8]. Existing recommender systems are often incapable of making recommendations precise and context-sensitive enough in this complicated environment.

Second, most traditional recommender systems, especially those that are based on association rule mining alone, can produce a wide range of recommendations that are made on the basis of overall popularity or co-occurrence patterns. Although helpful, these approaches might not provide the level of granularity or specificity needed to create really personalized learning paths. The heterogeneity of student behavioral patterns and learning preferences requires the existence of a mechanism that would help to refine and filter these general recommendations to fit better the specific characteristics of students and their learning objectives.

Lastly, the incorporation of sophisticated methods such as link prediction to recommender systems has a lot of potential in detecting subtle connections and similarities among users. The computational complexity of traditional link prediction algorithms, especially in the case of large-scale networks or large amounts of data that are characteristic of online education systems, however, makes direct application of such algorithms impractical. The situation is urgent to develop methodologies that will be able to utilize the power of link prediction with the reduction of its computational complexity.

To overcome these issues, the present paper suggests an APRIORI and Link Prediction-based Course Recommendation System (ALP-CRS). We are inspired to resolve these limitations by providing a new solution that not only detects the common patterns of behavior but also improves recommendations by intelligent combination of graph analysis, clustering and link prediction, thus providing a more accurate and personalized learning experience to students in online education.

1.2. Research Objectives and Contributions

To overcome the above-mentioned challenges, this paper introduces a new Personalized Recommender System of Courses in Online Education Systems called ALP-CRS, which combines association rule mining with sophisticated graph analysis methods. This work has the following contributions in overview:

- **New Hybrid Methodology:** We suggest a new hybrid approach that will mix the advantages of APRIORI-based association rule mining and graph clustering and link prediction methods. This combination gives a more accurate and individualized recommendation system to online education systems, where general trends are no longer sufficient, and highly relevant course recommendations should be given.

- **Improved Recommendation Precision by filtering rules:** Our system employs the elaborate study of the extracted association rules along with filtered similarities of students. This approach helps the system to narrow down the possibly wide suggestions based on the behavioral patterns, which raises the accuracy of the suggested courses considerably.
- **Computational Efficiency through Clustering and Link Prediction Integration:** We present a new integration of link prediction and clustering methods that are aimed at alleviating the computational complexities of link prediction algorithms. Our method reduces large networks into smaller sub-problems (clusters) and the application of link prediction can be efficiently applied, which is possible in the case of large datasets in online education.
- **Personalized Recommendation Domain Restriction:** The suggested approach uses graph analysis, clustering and link prediction to successfully reduce the recommendation domain. This strategic filtering makes sure that the system provides a very personalized set of courses based on the specifics of the individual student and the habits of his/her closest counterparts.

The paper is continued in this paragraph. Our study's second section focused on relevant studies. The third segment covered the tools and methods. A conclusion is drawn in the fifth part after a study of the data in the fourth section.

2. Related Research

The development of online education has seen the need to change the Learning Management Systems (LMS) to be a mere content repository to an intelligent and adaptive environment. The contemporary LMS systems have serious issues with scalability and the capacity to deliver individualized learning opportunities in the environment of increasing data volumes, as it has been emphasized in an in-depth analysis by [9]. This has led to the need to integrate modern recommender systems in these platforms as a key developmental concern to overcome these infrastructural challenges. It is in this backdrop that this research has been concerned with the analysis of the growing body of research on the customized RSs that are specifically targeted towards the purposes of e-learning.

To propose courses for learners, Obeidat et al. [10] and others employed the collaborative filtering approach and comparison of association criteria, which are based on classic data mining techniques. According to the experimental data, learners may greatly increase the recommendation impact by grouping and clustering.

Huang et al. [11] suggested assignments for students based on the Markov decision process and the reinforcement learning approach. They took into account the degree of learners' engagement, review and preview, and the smoothness of the exercise difficulty while making recommendations.

To give learners with tailored learning path recommendations, Liu et al. [12] suggested tracking learners' knowledge levels using a neural network-based technique.

The recommendation system for electronic books presented by Núñez-Valdez et al. [13] was based on user interaction with content. It tracked user behavior using indicators, fed a recommendation engine, and provided explicit ratings.

In their comprehensive review of machine learning-based RSs for e-learning, Khanal et al. [14] emphasized four traditional techniques: collaborative filtering, content-based, knowledge-based, and hybrid approaches. They looked at component needs, algorithms, input and output characterization, and assessment methodologies. In a survey of e-learning RSs, Zhang et al. [15] noted both established methods and novel approaches. For the construction of e-learning RSs, they suggested a structure that included a database server, recommendation engine, and user interface.

Using Kohonen-based unsupervised deep learning and a deep auto-

encoder, Mawane et al. [16] proposed a two-step model for effective resource suggestion. By locating helpful resources and managing high-dimensional data, this method improved the efficiency of learning resources.

Chen et al. [17] introduced a novel Click-Through Rate (CTR) model for customized online course selection in an effort to improve feature representation and solve the issue of cold start. It recorded users' cognitive habits using eye-tracking equipment and used transfer learning to address the data shortage. Ullah et al. [18] introduced a Deep Neural Collaborative Filtering method called Deep Edu. To increase the precision of recommendations for educational services, it learns N-dimensional and non-linear connections between users and books. Sivaramakrishnan et al. [19] suggested a Hybrid Bayesian Stacked Auto-Denoising Encoder (HBSADE) model to enhance the performance of recommender systems. To handle the nonlinearities in user and item interactions, it used a multilayer neural network, and to determine the user's interests, it used matrix factorization. Tests were carried out to show HBSADE's effectiveness. A deep learning-based course recommender system was described by Dien et al. [20] using pre-processing methods and Multi-Layer Perceptron. The system suggested courses based on its ability to forecast grades from four-million-mark records at Vietnam's Can Tho University. Positive results from the recommended strategy demonstrated that it may work in practical settings. A unique deep learning-based multi-criteria collaborative filtering approach was introduced by Nassar et al. [21]. For forecasting criterion ratings and total ratings, the model has two branches. The model fared better in the experiments than other approaches for every assessment metric.

In order to provide reliable and individualized suggestions for online courses, Wang et al. [22] used a learner model that included emotional components and a deep neural factorization machine. The experimental findings demonstrated that Deep-FM outperformed FM using the same model in terms of accuracy, recommendation accuracy rate, and root mean square error as well as average absolute error. This demonstrated how crucial learners' emotions are to their preferred methods of learning.

Tanwar and Vishwakarma [23] introduced a hybrid recommender system built on deep neural networks as an enhancement over current hybrid systems and traditional Collaborative Filtering. Training on high-dimensionality non-linear data using user and item vectors improved performance on publicly available datasets and yielded more precise suggestions. A Personalized Course Recommender System (PCRS) with hybrid approach and ontology was introduced by Gulzar et al. [24] to help students choose the right courses for their research projects. This method might mitigate the shortcomings of simple individual recommender systems, increasing performance and satisfaction levels and possibly improving learners' performance.

Learning efficiency was increased by the innovative DEep Learning Based Course Recommender system (DECOR) created by Li and Kim [25]. This system logged user activities and course characteristics well, addressing both data sparsity and information overload. Tests conducted on real-world datasets demonstrated that DECOR outperformed traditional methods and provided superior recommendations.

Nevertheless, although these hybrid structures are effective in integrating methods to enhance overall accuracy, they tend to use optimization on a global scale on the entire dataset. Conversely, our ALP-CRS solution is based on incremental innovation, involving the use of link prediction in localized and similarity-based clusters. This is a critical difference; as shown in the research [8], such granular improvements have to be expressed in relation to existing models to progress the student performance prediction. Our approach, which substitutes the global rule mining with the localized refinement, considers the particular behavioral subtleties that are normally overlooked by conventional hybrid models.

Recently, studies have been done to incorporate cognitive and contextual elements to increase similarity calculations in collaborative filtering. As an example, Jain et al. [26] introduced a Cognitive

Similarity (CgS) measure, which enhances the collaborative filtering performance by adding cognitive features to the rating information in order to compute user similarity. This method proved to be more accurate with a lower value of MAE and RMSE. Although CgS successfully applies contextual data in order to improve collaborative filtering, our proposed ALP-CRS applies a different approach. This is a multi-stage combination of data mining to extract patterns and graph-theoretic approaches to identify refined candidates, especially the novel application of clustering to address the complexity of link prediction in large educational data, which makes ALP-CRS unique and provides a different route to highly personalized course recommendations in online learning.

The recent development in educational technologies has seen a growing use of artificial intelligence and machine learning to streamline different academic and advising operations. As an example, deep learning-based intelligent advising systems, e.g., LSTMs, have been suggested to track student progress in online education and suggest student-specific interventions to complete courses on time [27]. In conjunction with student advising, there has been a major push in the semantic analysis of course materials and credentials to automate course equivalency and credit transfer systems [28]. To do this, recent research has considered more advanced Natural Language Processing (NLP) methods, including Word Mover's Distance (WMD), to measure the deep semantic similarity of course descriptions and learning outcomes in various institutions, frequently comparing the techniques with more traditional surface-level measures [29,30]. Though these NLP-based strategies are mainly concerned with cross-institutional credentialing and linguistic semantics, our suggested ALP-CRS is more concerned with intra-institutional behavioral patterns. Through data mining and graph-based link prediction, ALP-CRS models student similarities using historical course selection data, as opposed to textual course descriptions, which provides a unique, behavior-based personalized recommendation mechanism. Table 1 lists a summary of the specifications of the reviewed studies.

Although the previous studies have contributed immensely to the development of course recommendation systems, the following research gaps are identified. Most of the researches mainly examine user and item characteristics without considering the impact of context or the specific infrastructural constraints of modern LMSs. Also, the cold start problem, where the system needs to provide recommendations for users who have had little or no activity in the system is still an issue. However, due to the nature of e-learning platforms, we also face the problem of data sparsity, which also restricts the applicability of traditional recommendation algorithms. Some of them use intricate deep learning models that need much computational power and knowledge. Furthermore, most systems tend to focus more on how accurate the results are rather than how granular and personalized the specific student clusters are. To overcome such limitations, our proposed model adds contextual factors, uses a combination of approaches to avoid the cold start issue, utilizes graphs to deal with the sparsity of the data, adopts a relatively simple architecture suitable for LMS integration, and finally focuses on the user's experience by providing personalized and diverse recommendations.

3. Proposed Method

This section presents an APRIORI and Link Prediction-based Course Recommendation System names (ALP-CRS). The proposed method for recommending suitable courses to students in an online education system utilizes a combination of graph analysis and data mining techniques. In this approach, efforts are made to extract common behavioral patterns in course selections through association rule mining. Furthermore, to narrow down the scope of recommendations and provide more accurate suggestions, graph clustering and link prediction techniques are employed. The proposed method can be summarized in the following phases:

Table 1

Summary of the specifications of the reviewed studies.

Reference	Year	Research Goal	Method	Limitation
Obeidat et al. [10]	2019	Recommend courses	Collaborative filtering and association rule mining	Limited to basic data mining techniques
Huang et al. [11]	2019	Recommend exercises	Markov decision process and reinforcement learning	Requires accurate user modeling
Liu et al. [12]	2019	Recommend learning paths	Neural network-based knowledge tracking	Complex model and data requirements
Núñez-Valdez et al. [13]	2018	Recommend eBooks	Implicit feedback-based recommendation	Relies heavily on user interactions
Khanal et al. [14]	2020	Review of machine learning-based RSs	Literature review	Does not propose a specific method
Zhang et al. [15]	2021	Survey of e-learning RSs	Survey	Does not propose a specific method
Mawane et al. [16]	2020	Recommend learning resources	Unsupervised deep learning	Requires large amounts of data
Chen et al. [17]	2020	Recommend online courses	Click-through rate prediction with eye-tracking data	Requires specialized hardware and data collection
Ullah et al. [18]	2020	Recommend educational services	Deep neural collaborative filtering	Complex model and computational cost
Sivaramakrishnan et al. [19]	2021	Improve recommendation performance	Hybrid Bayesian Stacked Auto-Denoising Encoder	Complex model and data requirements
Dien et al. [20]	2020	Recommend courses	Deep learning-based prediction model	Relies heavily on accurate grade data
Nassar et al. [21]	2020	Multi-criteria recommendation	Deep multi-criteria collaborative filtering	Requires accurate rating data
Wang et al. [22]	2022	Personalized course recommendation	Deep neural factorization machine with emotional factors	Requires accurate emotion detection
Tanwar and Vishwakarma [23]	2023	Hybrid recommender system	Deep neural network-based hybrid approach	Complex model and data requirements
Gulzar et al. [24]	2018	Course recommendation for research projects	Hybrid approach with ontology	Limited to research project context
Li and Kim [25]	2021	Course recommendation	Deep learning-based course recommender system	Requires accurate user activity data
Jian et al. [26]	2022	Enhance CF-based RS performance	CgS, integrating contextual features with ratings for improved similarity	Less focused on complex behavioral patterns beyond direct user-item interaction.

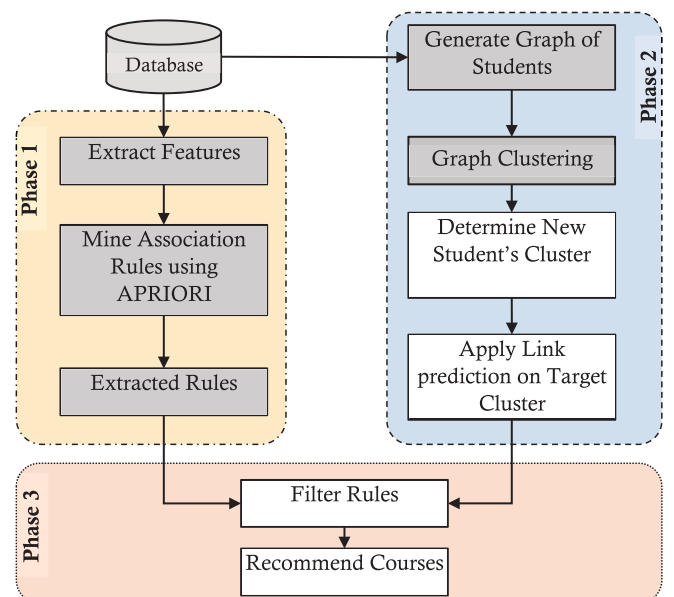
1. An explanation of the guidelines for choosing a course: This phase's goal is to find common trends in the courses that students choose to take, and it does this by using the following comprehensive steps:
 - Feature extraction from the course selection database
 - Extraction of rules related to course selection
2. Analysis of student features based on similarity: In this phase, the set of individual and behavioral characteristics of users is analyzed using graph analysis techniques to restrict the recommendation domain for more precise suggestions. This process is conducted through the following detailed steps:
 - Building and clustering student similarity graphs
 - Filtering similar students based on link prediction
3. Recommendation delivery: In this phase, the extracted patterns from the first phase are filtered based on rules corresponding to identify candidates in the second phase, and the system's output recommendations are generated based on these filtering results.

It should be mentioned that in contrast to traditional systems based on association-rules, which tend to produce overly general rules with high false-positive rates, ALP-CRS uses a hybrid structure. The proposed method filters general association rules with the help of specific student similarities by incorporating graph-based link prediction. This method is in line with the recent comparative models [6], which focus on the fact that the integration of data mining with particular user-profile analysis is a far more accurate method of classification and reduction of error.

The accuracy improvement of ALP-CRS is done by a two-layered filtering approach which switches between the global and local relevance. The APRIORI algorithm finds high-probability transitions between courses in the entire student population in the first layer, which gives a powerful yet wide set of candidate rules. Nonetheless, such general rules tend to decrease the precision since they fail to consider the sub-contexts of individual students. The second layer tackles this with the help of graph clustering and link prediction to establish a high-fidelity neighborly context of the target student. The system effectively removes noisy or irrelevant recommendations by weighing up the general association rules against the actual behavior of these similar peers (as formulated in Equation (5)). This composite methodology has the advantage of making sure that a course is recommended when it not

only is statistically significant within the overall dataset but also is quite relevant to the learning community of the student in question so as to optimize the accuracy and saliency of the eventual output.

Moreover, to guarantee the strength of the findings and the error reduction that is recommended in [31], the performance of the proposed model was tested on the basis of a k-fold cross-validation protocol, the specifics of which are presented in the experimental section. The proposed mechanism for providing recommendations is illustrated in Fig. 1. In this figure, the steps associated with each phase of the proposed method are separated. Given the proposed structure, the first and second phases can be executed in parallel due to their data independence. The computational processes of a recommender system can be performed offline (building the recommender model based on data history) and online (providing recommendations for new instances). In Fig. 1, offline

**Fig. 1.** Proposed mechanism for providing recommendations.

processes are depicted as gray blocks and online processes as white blocks. The computational details of the proposed method are explained in the following sections of this paper. Table 2 represents the list of notations used in this paper.

The rest of this section is dedicated to describing each step of the proposed approach, in details.

3.1. Describing Course Selection Rules

The first phase in the proposed method is the extraction of course selection patterns in the online education system based on students' performance records. This phase begins with the extraction of features from the database transaction set and ends with the extraction of association rules using the APRIORI algorithm, which is explained in the following sections.

a) Feature Extraction from the Course Selection Database

The first step in the phase of describing the proposed rules is to convert the raw and unprocessed data in the database into a format that can be processed by subsequent steps. Initially, a database containing course selection information for students is considered. This database can include information such as student details, courses selected by students each term, and also course specifications. The aim of the feature extraction step in the proposed method is to eliminate redundant information and extract course selection information related to each student. For this purpose, first, considering the relationships between the student, course selection, and course tables, the course selection records of each student in different terms are extracted. By doing this for each student, a structure as shown in Table 3 is obtained.

In the above structure, Student ID is an identifier for students, Course ID is an identifier to determine the students' field of study, and Section indicates the term number for which the course selection information for a student is available. Also, in the displayed structure, $d_{i,j}$ represents the i -th course selected by a student in their j -th term of study. In the next step, the courses selected by students are converted into transactions of variable length. For this purpose, first, the structures obtained for students are filtered based on the field of study. That is, only the information related to students studying in the field of interest is considered. Then, the course selection information of students in consecutive terms is considered as transactions that form the features extracted from the database. This structure is formed for all consecutive terms related to the course selection of students currently studying in the field of interest and is used as input to the APRIORI algorithm for the extraction of association rules.

b) Extraction of Rules Related to Course Selection

After extracting features from the course selection database of students, the APRIORI algorithm is used to extract patterns of course selection by students. Each record in the database resulting from the previous step is considered as a transaction. The APRIORI algorithm

Table 2
List of notations.

Notation	Description
T	A transactional database containing the selected courses.
C_k	Set of candidates at level k in the APRIORI
L_k	The collection of extracted rules that are transmitted as the APRIORI output at level k
x_i	The i -th attribute in demographic structure of student x
n	Number of attributes describing a student
N	Number of students in the dataset
$score_{v_y}$	he degree centrality of vertex v_y
v_x	A vertex corresponding to student x in graph of similarities
$L(v_x)$	The set of vertices that have a connection with a minimum length of 2 with vertex v_x
$path_{v_x, v_y}^i$	The collection of paths without a cycle and a length of i between vertices v_x and v_y
$sim(v_x, s_i)$	The similarity between the student x and the i -th candidate student from set S

Table 3

Database to store information on students' course selection.

<Student ID>	<course ID>
Section1	$d_{1,1}, d_{2,1}, d_{3,1}, \dots$
Section2	$d_{1,2}, d_{2,2}, d_{3,2}, \dots$
Section3	$d_{1,3}, d_{2,3}, d_{3,3}, \dots$
...	...

derives a set of principles by examining transactions for patterns. The classic algorithm for learning dependency principles is APRIORI. Databases containing transactions (sets of courses selected by students in consecutive terms) form the basis of this algorithm. There are additional algorithms in this domain that operate on databases devoid of transactional information. This algorithm accepts as input a collection of item sets. The algorithm attempts to find subsets of items that are at least common among C item sets. APRIORI is a bottom-up algorithm, such that at each stage an item is added to the recurring subsets (candidate generation). The candidate sets are evaluated on the data. The termination condition of the algorithm is the absence of any other successful development method. The output of the APRIORI algorithm, which seeks to identify dependencies between distinct data sets, is a set of principles describing how items are incorporated into data sets. For the efficient enumeration of item sets, APRIORI employs first-level search and a tree-like model. The algorithm for a transactional database T and a threshold value ϵ is represented by the following pseudocode [32].

Algorithm1: ARIORI Algorithm

```

 $L_1 \leftarrow \{\text{large 1-itemsets}\}$ 
 $k \leftarrow 2$ 
While  $L_{k-1} \neq \emptyset$ 
   $C_k \leftarrow \{c | c = a \cup \{b\} \wedge a \in L_{k-1} \wedge b \in \bigcup L_{k-1} \wedge b \not\subseteq a\}$ 
  For transactions  $t \in T$ 
     $C_t \leftarrow \{c | c \in C_k \wedge c \subseteq t\}$ 
  For candidates  $c \in C_t$ 
     $\text{Count}[c] \leftarrow \text{Count}[c] + 1$ 
   $L_k \leftarrow \{c | c \in C_k \wedge \text{count}[c] \geq \epsilon\}$ 
   $k \leftarrow k + 1$ 
Return  $\bigcup_k L_k$ 

```

The algorithm described above employs symbols from standard set theory. Consider T to be a multiset. Level k 's candidate set is denoted by C_k . Additionally; L_k represents the collection of extracted rules that are transmitted as the algorithm's output. L_1 comprises all items that are present in a minimum of one hundred percent of transactions. L_{k-1} is utilized to acquire L_k , just as L_1 is utilized to acquire L_2 . "All non-empty subsets of a large item set must be large item sets" is a crucial function that aids in the generation of L_k and increases the algorithm's efficacy. The validity of this proposition is easily observable, according to the definition of an item set like A that does not have a minimum Support is an unimportant set, if another item (or items) like X is added to A , the resulting set, i.e., $A \cup \{X\}$, certainly cannot be an important set. The steps to generate L_k s are briefly as follows. Initially, to obtain L_k s, a set of k -item sets is generated by combining members of L_{k-1} as candidates, which are called C_k . The combination $L_{k-1} * L_{k-1}$ is done when the members of L_{k-1} have exactly $k-2$ items in common, or in other words [32]:

$$L_{k-1} * L_{k-1} = \{A * B | A, B \in L_{k-1}, |A \cap B| = k - 2\} \quad (1)$$

In Equation (1), L_{k-1} is the collection of frequent itemsets of length $k-1$. The join operation is used to merge two sets when the first $k-2$ elements are the same and it is used to ensure that candidate sets C_k are generated efficiently. Following this is the pruning phase. Although C_k contains every element of L_k , not every element of C_k is necessarily significant. Although a solitary examination of the databases can detect critical entries, an improved resolution also exists. The APRIORI property is employed to accomplish this; that is to say, if any of the $k-1$

subsets of C_k do not appear in L_{k-1} , then C_k cannot possibly be an essential set. Database visits are restricted to specific members of C_k who meet the specified conditions. Once the significant item sets have been identified, the subsequent course of action is to identify strong association rules from this set. Such rules are those that meet the following condition [32]:

$$\text{confidence}(A \Rightarrow B) = P(B|A) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)} \geq \text{min} - \text{conf} \quad (2)$$

In this case, A and B are disjoint sets of courses with the rule being $A \Rightarrow B$. The term $\text{Support}(A \cup B)$ is the fraction of transactions in database T that include both sets A and B and $\text{Support}(A)$ is the frequency of the antecedent set A . On the basis of this relationship, the following principles are generated: All non-empty subsets of an important item set I are generated, and if the confidence level for any of the subsets s is equal to or greater than a predetermined threshold, that rule is incorporated into the set of rules. It should be noted that the support values for each of the important sets and their subsets have already been calculated, eliminating the necessity for recalculation, given that the important sets have already been generated. In the implementation of this phase, to balance rule quantity and quality, the minimum support threshold (epsilon) was set to 0.2 and the minimum confidence threshold was set to 0.6.

3.2. Student Feature Analysis Based on Similarity

Graph analysis techniques are employed in the second phase of the proposed method to examine student characteristics in order to restrict the recommendation domain and generate more accurate suggestions. When determining which resources are most appropriate for a particular student, it is possible to draw upon the prevailing behavioral patterns exhibited by similar students. Link prediction can function as a highly effective approach to accomplishing this objective. Nonetheless, the considerable intricacy of this approach renders its implementation unfeasible on expansive datasets. By employing clustering techniques, the problem's complexity can be diminished. In line with the transparent methodological reporting requirements of educational recommender systems [4], the concrete criteria of graph construction and clustering is described below. The database describing students is initially transformed into a graph for this purpose; subsequently, this graph is clustered into a collection of groups according to similarities among students. Subsequently, the algorithm utilizes the link prediction technique to derive a set of candidate students for recommendation pattern extraction from the cluster containing the target student. In order to accelerate the execution of this phase in the proposed method, parallel processing techniques are utilized, considering the data independence of this phase from the initial phase.

a) Constructing a Student Graph and Clustering

Commencing the second phase of the proposed methodology is a graphical description of the students' database. In this case, the student graph is a user-user similarity graph $G = (V, E)$ where V is the set of students and E is weighted edges which are weighted based on the behavioral similarity. This contrasts with bipartite user-course graphs frequently utilized in collaborative filtering; in this case, direct relationships are the closeness between the demographic and academic profiles of two students. During this stage, every student is represented as a vertex of the graph, with each edge signifying a significant similarity between the characteristics of students enrolled in the same academic discipline. Therefore, in order to establish the aforementioned graph structure, an edge is initially regarded for each pair of students, denoted as i and j , who are pursuing the same field of study. This process results in the database being described in the form of a set of complete sub-graphs, where each sub-graph represents students belonging to one of the fields of study. These complete sub-graphs only describe the field of study similarity of students through their links and do not provide information about the similarity of student features. Subsequently, for

each sub-graph, the pairwise similarity among its members is calculated in the form of the cosine similarity metric, and this metric is used to weight the connections of each sub-graph [33].

$$W(x, y) = \frac{\sum_{i=1}^n x_i y_i}{\sqrt{\sum_{i=1}^n x_i^2} \sqrt{\sum_{i=1}^n y_i^2}} \quad (3)$$

In Equation (3), $W(x, y)$ is the similarity coefficient between students x and y . The x_i and y_i are the numerical values of the i -th attribute of students x and y , respectively. In addition, n is the number of dimensions (demographic and academic attributes) of the feature vector. The numerator ($\sum_{i=1}^n x_i y_i$) computes the dot product which is the alignment of the two vectors. Also the denominator ($\sqrt{\sum_{i=1}^n x_i^2} \sqrt{\sum_{i=1}^n y_i^2}$) of the product of the Euclidean magnitudes normalizes the result to lie within the range of 0 to 1.

Although it is true that cosine similarity, as any metric, has its limitations (e.g., sensitivity to sparse data, incapability to reflect the difference in magnitude when only direction is relevant), it is chosen in this study because it has several important strengths that fit our research aims and the characteristics of our data. To begin with, cosine similarity is really good at gauging the direction of two vectors as opposed to their length. This property is especially relevant in a comparison of student feature vectors in which the pattern or relative proportion of attributes (e.g., GPA, number of courses completed, or frequency of selection of particular courses) is frequently more suggestive of behavioral and characteristic similarity than the absolute values. As an example, two students with very different GPAs but who take courses in similar ways (e.g., always do well in quantitative courses) may be said to be more similar in learning style, and cosine similarity adequately reflects this directional similarity. Second, the cosine similarity is naturally scale-invariant to feature values. This is a big plus in the case of combining different student characteristics that inherently have different numerical ranges (e.g., total GPA vs. number of withdrawn courses). Implicitly normalizing these attributes by calculating the cosine similarity helps to avoid the situation when features with higher numerical values contribute disproportionately to the similarity score, and all features make meaningful contributions to the similarity measure. Lastly, the computational efficiency of cosine similarity is also in line with our objective of dealing with complexity of the subsequent link prediction operations, particularly following the initial clustering which brings down the dimensionality. It is an effective option in terms of large-scale similarity calculations in the graph construction step, which adds to the overall practicability and efficiency of our proposed ALP-CRS system because of its simple computation.

Based on the calculated weight values, the subsequent phase of the proposed method is to decompose each subgraph into a collection of communities. To accomplish this, a graph may be described using the tree structure described in [34]. By virtue of the tree structure, every vertex of the graph is evaluated a single time. Consequently, the computational intricacy of the suggested approach is diminished. In order to discern communities via the constructed weighted graph, the maximum spanning tree associated with the weighted graph is initially generated utilizing the Prim algorithm. This operation results in the elimination of certain connections that have a low weight in the graph. The Prim algorithm generates the maximum spanning tree through the subsequent steps:

Algorithm2: Maximum Spanning Tree Algorithm

1. **Input:** A weighted connected graph with vertices (V) and edges (E).
 2. **Initialization:**
 - Initialize $V_{\text{new}} = \{x\}$ with a single arbitrary vertex (x).
 - Set $E_{\text{new}} = \{\}$ to an empty set.
 3. **Repeat until** ($V_{\text{new}} = V$):
 - Select an edge (u, v) with maximum weight such that (u) is in V_{new} but v is not.
 - Add vertex (v) to V_{new} and edge (u, v) to E_{new} .
 4. **Output:** The resulting V_{new} and E_{new} form the maximum spanning tree.
-

After forming the maximum spanning tree, the next step is to remove the edges with the highest weight in the obtained tree. If the graph has N vertices, it is possible to remove $\left\lfloor \frac{N-1}{2} \right\rfloor$ connections with the highest

weight in the obtained maximum spanning tree to form $\left\lfloor \frac{N+1}{2} \right\rfloor$ local communities. Students residing in each of these local communities exhibit the greatest degree of similarity in both characteristics and behavior, owing to the fact that their internal edge connections carry the most weight. It should be noted for nodes without any connections with others (for example new student), a hypothetical link between the student and the student having maximum similarity in demographic attributes (field of study, academic semester, gender, location) is assumed. In other words, for new users an edge between him/her and the most similar user is considered which makes the user a member of the cluster containing most similar students. This approach can effectively handle the problem of cold start for new users.

In the second phase of the proposed method, which identifies candidate nodes for recommendation by applying the link prediction technique to the members of each of the obtained local communities, these communities serve as input.

b) Filtering Similar Students Based on Link Prediction

The link prediction algorithm is applied to identify filtered comparable students, i.e., a subset of users in the same cluster that has the highest calculated similarity scores with the target user. The recommender system's online phases consist of this step and the ones that follow in the suggested manner. This indicates that suggestions for new instances are generated by way of these procedures. The basic link prediction algorithm considers communication paths of variable lengths between the vertices of the graph and performs link prediction based on that. The link prediction algorithm includes two phases of processing connections and calculating similarity. In the connection processing phase, all connections between vertices are described in the form of a matrix containing single-step connections (paths with a length of one). Accordingly, if there is a connection from v_i to v_j between two vertices like v_i and v_j in the graph, then in the row v_i and column v_j of the matrix, the identifier corresponding to the destination node of the connection (i.e., v_j) is stored. This matrix can display all paths resulting from connections with a length of one between vertices. In the similarity calculation phase, the created single-step path matrix is used to generate paths with longer lengths. For example, by combining two paths with a length of one, a path with a length of two can be created.

In order to combine two paths and generate paths with a greater length, all graph connections must be examined through three nested loops. With the increase in path length (l), this complexity increases.

Traditional link prediction on a global graph has a third-degree computational complexity, $O(N^3)$ where N is the total number of students. This renders typical link prediction very inefficient and likely to have serious slowdowns as the user base and course catalog increase. To eliminate this scalability bottleneck, the suggested ALP-CRS architecture mathematically limits the execution time by dividing the global graph into local clusters. In case the graph is partitioned into k clusters of average size n ($n = N/k$), the link prediction is only performed in these closed communities. This means that the complexity is no longer a global $O(N^3)$, but a local sum of $k \cdot O(n^3)$. Since n is tightly constrained and much less than N , the total growth rate of the computational cost is severely decreased to $O(\frac{N^3}{k^2})$. This mathematical reduction is necessary to make sure that the system does not slow down exponentially with the addition of new users. Moreover, the practical implementation of this system, as explained later in Section 5.2, is based on the computation of these $O(N^3)$ matrices offline so that the online delivery of the recommendations can be instantaneous to the end-user. The link prediction algorithm makes recommendations based on the sub-graph clusters.

Based on this, the cluster containing the target student is subjected to the link prediction method, which creates a similarity matrix of vertices based on the connections between them.

The last stage involves recommending a group of students based on the algorithm's output who have the greatest similarity score to the target student. For this reason, the similarity between the vertices should be able to be calculated once pathways of length l are generated. The suggested technique creates a similarity matrix of vertex connections by combining two centrality and route similarity measures. The similarity matrix is updated in order to compute the similarity of students using the link prediction method, which is based on pathways of varying lengths that exist between vertices. Assuming that the graph has no repeated vertices, a matrix of routes with lengths of two and three may be created if there are two vertices, v_x and v_y . The likelihood of resemblance increases with the number of these pathways. Therefore, assuming a matrix containing paths with lengths one and two for all pairs of graph vertices, the similarity between two vertices v_x and v_y can be calculated as follows:

$$\text{sim}(v_x, v_y) = \frac{\text{score}_{v_y}}{\sum_{v_z \in L(v_x)} \text{score}_{v_z}} + \sum_{i=2}^l \frac{1}{i-1} \cdot \frac{|\text{path}_{v_x, v_y}^i|}{\prod_{j=2}^i (n-j)} \quad (4)$$

Where n represents the number of vertices in the graph, and l is the maximum length considered for a path between two vertices v_x and v_y . The maximum path length was determined to be $l = 3$ in this study based on initial experiments to trade off between the cost of computation and the accuracy of prediction. $L(v_x)$ denotes the set of neighbor-of-neighbor of v_x . i is also the length of a current path under consideration (between 2 and l), and j is an index variable of the product operator ($\prod$) to compute the number of combinations of paths in a graph of size n (number of vertices). Also, score_{v_y} indicates the degree centrality of vertex v_y , and $z \in L(v_x)$ identifies the set of vertices that have a connection with a minimum length of 2 with vertex v_x . In these paths, there cannot be a cycle. $\frac{1}{i-1}$ is a decay factor that weights paths of different lengths. For example, paths with a length of two are considered with a weight of $\frac{1}{2-1} = 1$, while paths with a length of three play a role in the similarity calculation relationship with a weight of $\frac{1}{3-1} = 0.5$. In the above relationship, $|\text{path}_{v_x, v_y}^i|$ is the total number of paths without a cycle and with a length of i between two vertices v_x and v_y .

A similarity matrix that forecasts the degree of similarity between the students in a cluster is produced after the application of the link prediction algorithm to the pupils within that cluster. Subsequently, the method of selecting comparable pupils is carried out by using the greatest likelihood in this matrix. Students that have the greatest values in the i -th row of the similarity matrix are therefore regarded as comparable to i . An example is shown to illustrate how comparable students are filtered. It is assumed that based on the clustering performed in the previous section, a subgraph corresponding to the structure shown in

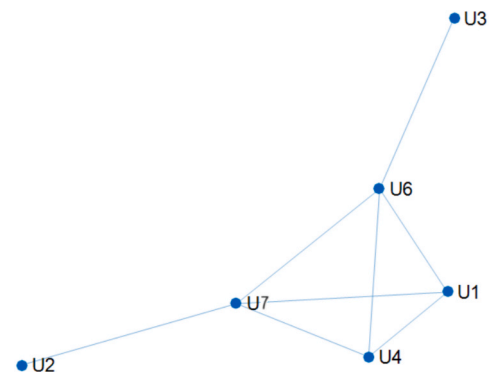


Fig. 2. Hypothetical graph for applying the link prediction algorithm.

Fig. 2 has been obtained, and the link prediction operation is performed for the vertex U3 located in this subgraph. In this figure, each student belonging to the obtained cluster has been presented by labels U1-U7.

If the maximum path length prediction parameter is considered to be $L = 3$, the matrix of paths between vertices is as Table 4. Also, the vertex similarity matrix based on eq. (4) is shown in Table 5. The indirect similarities between the vertices are represented by the matrix in Table 5. The vertices that have a direct connection to the target student are added to the list of vertices that have the greatest similarity based on link prediction at the conclusion of this phase. The third stage of the suggested technique uses this set as input to carry out the rule filtering operation and provide suggestions based on the results.

3.2.1. Providing Recommendations

The third stage of the suggested approach takes two sets as input: “Similar Students S” and “Extracted Rules R.” These sets are used to combine rules based on comparable students’ past behavior to execute suggestions. This means that if a rule such as r_i corresponds with the conditions of one of the members of set S, in addition to the target student’s criteria, it serves as the foundation for a suggestion. To suggest a new item to students, first, the courses previously selected by the student are extracted, and then the courses that are members of this set are searched in the left side of the filtered rules. Any rule that has the previously selected courses of the student on its left side can be used to suggest courses to the student. Subsequently, this set of recommendable selections for student x is ranked as follows:

$$F = \sum_{i=1}^{|S|} \sum_{j=1}^{|R|} \delta_{ij} \times \text{sim}(v_x, s_i) \times \text{confidence}(r_j) \quad (5)$$

Where $\text{sim}(v_x, s_i)$ represents the similarity between the target student x and the i-th candidate student from set S, which is calculated through the link prediction algorithm in the second phase of the proposed method. Also, $\text{confidence}(r_j)$ indicates the confidence level of the j-th usable rule in set R. Ultimately, δ_{ij} is a binary function and is only equal to one if the course selection history of student s_i matches the constraints in rule r_j . $|S|$ and $|R|$ are the total number of students in the similarity set which underwent a filtering in the last ranking formula, and the total number of candidate association rules, respectively. The indices i and j go through these sets to sum the weighted contribution of all similar peers behavior to the final recommendation score F. After ranking the

Table 4
Path matrix between vertices for the graph of Fig. 2.

	U1	U2	U3	U4	U6	U7
U1		U1 → U7 → U2	U1 → U6 → U3	U1 → U4	U1 → U6	U1 → U7
U2	U2 → U7 → U1 U2 → U7 → U4 → U1 U2 → U7 → U6 → U1		U2 → U7 → U6 → U3	U2 → U7 → U4 U2 → U7 → U1 → U4 U2 → U7 → U6 → U4	U2 → U7 → U6 U2 → U7 → U1 → U6 U2 → U7 → U4 → U6	U2 → U7
U3	U3 → U6 → U1 U3 → U6 → U4 → U1 U3 → U6 → U7 → U1	U3 → U6 → U7 → U2		U3 → U6 → U4 U3 → U6 → U1 → U4 U3 → U6 → U7 → U4	U3 → U6	U3 → U6 → U7 U3 → U6 → U1 → U7 U3 → U6 → U4 → U7
U4	U4 → U1	U4 → U7 → U2	U4 → U6 → U3		U4 → U6	U4 → U7
U6	U6 → U1	U6 → U7 → U2	U6 → U3	U6 → U4		U6 → U7
U7	U7 → U1	U7 → U2	U7 → U6 → U3	U7 → U4	U7 → U6	

Table 5

Similarity matrix between vertices based on centrality-based link prediction.

Method	Mean Reciprocal Rank (MRR)	Catalog Coverage (%)
APRIORI	0.5216	34.17
PCRS [24]	0.5834	49.56
DECOR [25]	0.6111	53.29
ALP-CRS (Proposed)	0.6807	58.13

recommendable selection set by eq. (5), the members of this set are presented to the student based on the resulting ranking.

3.3. Reliability and Security Analysis

In order to make sure that ALP-CRS can be practically applied in institutional settings, the system includes certain mechanisms aimed at the operational reliability and information protection.

3.3.1. System Reliability

The three main mechanisms that ensure the reliability of ALP-CRS are:

- **Algorithmic Determinism:** The community detection algorithm makes use of the Prim algorithm (Algorithm 2) to build the Maximum Spanning Tree. Prim is deterministic, unlike stochastic clustering algorithms; that is, given a fixed set of similarity weights, the clusters of students obtained by Prim are the same ones obtained by repeated executions.
- **Stability through Decay Factors:** The decay factor ($\frac{1}{i-1}$) in Equation (4) makes the system give preference to direct connections that are stable. The model avoids the problem of recommendation drift by punishing longer and less speculative trajectories and preserves high-confidence recommendations.
- **Robustness to Cold Start:** To ensure the system is reliable to new users, it uses a demographic mapping approach (Section 3.2.1) whereby users who have no history are assigned to the closest existing cluster. This will make the system functional and reliable even at the early stages of student enrolment.

3.3.2. Data Security and Privacy

The issue of security in ALP-CRS is addressed by a Security-by-Design method of safeguarding sensitive student records:

- **Data De-identification:** Prior to the graph construction, a layer of data anonymization will replace unique Student IDs by synthetic ones. This eliminates the possibility of re-identifying people based on the analysis of graph topology.
- **Role-Based Access Control (RBAC):** The architecture limits the access to the raw feature database to the internal backend of the recommendation engine. The final, filtered recommendations are only forwarded to the user interface, and sensitive demographic and academic information is not exposed to external parties or unauthorized users.

4. Research Finding

The suggested ALP-CRS was tested on a 10-fold temporal cross-validation protocol to guarantee the reliability, strength, and external validity of the results. To recreate a real-world recommendation situation, the dataset was divided into 10 different segments according to the time aspect (chronological order of semesters). The model was trained in each fold of the validation on the historical course selection data of the previous intervals and tested on the following unseen interval. This sliding window methodology guarantees that the model is tested on its predictive capability of future choices based on historical behavior, which is a good way of avoiding data leakage and over-fitting. In the

continuation of this section, after describing the database specifications, the evaluation metrics and research findings are presented.

4.1. Data Collection

The dataset used in this research includes information on 13,344 students studying in various fields at Ilam Islamic Azad university, Iran. This data is stored in a MySQL 5.5 database. In this database, course selection information for 42,581 terms related to these students is stored. To follow ethical guidelines in employing these data, a subset of 367 available students for obtaining written informed consent were selected from this dataset. The resulting subset of data, focuses on recommending courses to master's degree students in computer science. In this set, information about computer science students and their course and resource selection has been extracted. After doing this, 738 course selection records related to 367 students over 10 academic terms were extracted. During the data cleaning process for the dataset, information about students whose course selection records for only one term were available in the database was ignored. Alongside the registration history information of students in the datasets there is a table describing student characteristics. In this table, each data record represents information about a student, which can be referred to via a foreign key to the table of their registered course history. Each data record in this table contains details on the student's year of admission to the faculty, area of study, and orientation, as well as their total GPA and the number of courses they successfully completed, withdrew from, and failed. Fig. 3 depicts the database structure utilized in this study.

In Fig. 3, the ST table contains personal information about students. The DR table stores the specifications of various courses. Also, the SL table stores information about students' course selections in different terms. Each of these datasets provides the necessary input for the proposed course recommender model. In order to feed this data to the ALP-CRS model, a three step pre-processing pipeline was adopted:

- 1) Data Cleaning: In addition to the removal of one-term records, we also processed missing GPA values with the departmental average of that cohort. The records that had inconsistent dates (e.g., graduation date before admission) were eliminated.
- 2) Normalization: Min-Max scaling was used to normalize numerical attributes (GPA and courses completed) to the range [1]. This is to avoid bias of the cosine similarity computation in Equation (3) by attributes with greater absolute numerical ranges.
- 3) Formatting of the course selection History: The course selection history was converted to a horizontal transactional format with each row containing a distinct (Student_ID, Term) pair and a comma-

separated list of Course IDs. This is the direct input of the APRIORI algorithm in Section 3.1.2.

It should be noted that all the 367 participants provided written informed consent prior to their participation. The study was conducted in accordance with ethical guidelines and regulations. The required steps were taken to ensure the privacy and well-being of participants by anonymizing all data and minimizing potential risks. Therefore, in this study the institutional licensing committee approval was waived due to the minimal risk posed to participants and the nature of the research, which involved a simple questionnaire and did not require invasive procedures or sensitive data collection.

4.2. Results

In implementing the proposed approach, the min-confidence threshold was considered as 0.65. Also, the path length for the link prediction algorithm was set as 3. The appropriate value for these parameters was determined experimentally. All the experiments were executed on a personal computer running Windows 11 64-bit on an Intel Core i7 13700H CPU and 16 gigabytes of RAM. Precision, recall, and F-measure were used as the assessment metrics after the recommendation operations were completed using these data and the outcomes compared with the actual courses that the students had selected. Additionally, there have been two distinct stages of experimentation. In the first phase, three metrics have been examined based on different numbers of repetitions, with only one recommendation being considered as the output for each sample. In the second stage, the suggested approach was contrasted with three other modes, and metrics were taken into account for variations in the quantity of suggestions. The suggested approach described in the third portion of the study is represented by the first mode. In the second mode, courses are recommended by using the APRIORI algorithm to extract rules and patterns. The third and fourth modes correspond to the PCRS [24] and DECOR [25] papers, which have been compared for evaluation.

The experiments were conducted in a well-organized five-stage working process to provide consistency. The initial phase was the graph construction, where the student similarity graph was constructed, using the demographic features (ST Table), and the weight of the edges was the calculated cosine similarity. The second phase was clustering and in that process, the Prim-based community detection was used to divide the graph into local clusters. Rule mining was conducted during the third stage. In order to achieve this, the APRIORI algorithm was applied to the transactional data in parallel to produce the global association rules (L_k). Stage 4 was devoted to the adjustment of parameters. Thus, we performed sensitivity tests to determine the best values of ϵ (support)

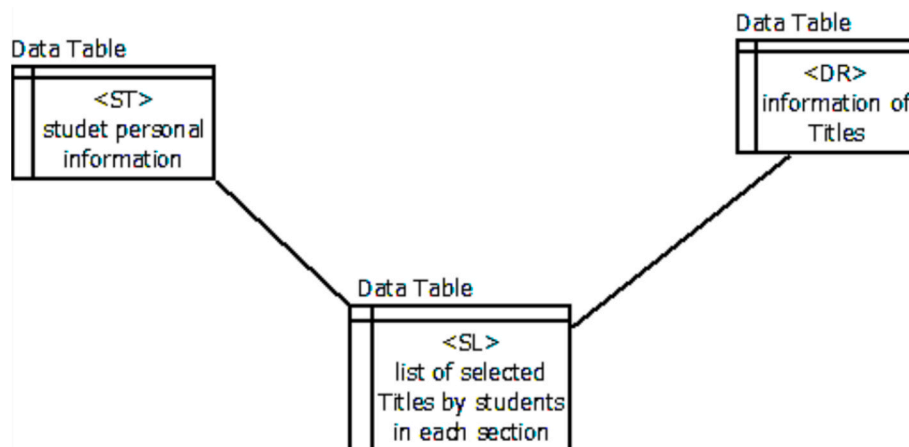


Fig. 3. Structure of the input database.

and confidence before the last run. Lastly, the 10-fold temporal cross-validation was implemented in the last step. Similarity of the target student to the students in the cluster in each of the folds was estimated using link prediction (Equation (4)) and the rules were filtered and ranked (Equation (5)) to generate the top-N recommendations.

In the context of a personalized recommender system for online education courses, each recommended item will have one of the following conditions.

- True Positives (TP): The courses recommended to a user by system which have actually been selected by that user also in the ground-truth set.
- True Negatives (TN): The courses that were not recommended by the system and also, have not been selected by the users in the ground-truth set.
- False Positives (FP): The courses recommended to a user which have not been selected by that user in the ground-truth set.
- False Negatives (FN): The number of courses that were not recommended by the system but have been selected by the user in the ground-truth set.

Precision refers to the ratio of recommended items that are genuinely preferred by users to the total recommended items by the system. True positives are the correctly recommended items that users like, while false positives are the incorrectly recommended items that users do not prefer, as shown in Eq. (6).

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

The number of true positives divided by the total of true positives and false negatives yields the recall value. False negatives show relevant products that were not advised, and true positives show the successfully recommended goods that match consumers' tastes. In Equation (7), the recall formula is shown.

$$Recall = \frac{TP}{FN + TP} \quad (7)$$

The F-measure is commonly defined as the harmonic mean of Precision and Recall values. The F-measure is calculated using Eq. (8).

$$F - Measure = \frac{2 * Precision * recall}{Precision + recall} \quad (8)$$

The effectiveness of the model was validated by comparing the generated recommendations with the ground truth that is the actual courses that the students took during the test period. The Precision, Recall and F-measure are calculated to give a quantitative estimate of the closeness of the system suggestions to the actual student requirements. Besides, to prove the superiority of ALP-CRS, the results were compared to baseline model (APRIORI-only) and two state-of-the-art approaches, PCRS [24] and DECOR [25], under the same experimental conditions.

Based on Fig. 4, the precision metric is displayed for different numbers of iterations. The important finding in these iterations is that, when compared to the APRIORI approach, the suggested method performs noticeably better. The graph shows how accuracy has been successfully boosted by using link prediction and clustering approaches. The results indicate that the proposed method has achieved a 10% increase in precision compared to using only the first phase (rule extraction). Furthermore, in comparison to the PCRS [24] and CEDOR [25] methods, it has consistently shown better performance in terms of precision across all iterations, demonstrating that the recommendations provided by the proposed method have higher precision compared to the other evaluated methods.

Fig. 5 illustrates the recall metric. The recall curve clearly indicates that the values of recall are also greater in comparison to the reference method. These findings show that our suggested technique is more

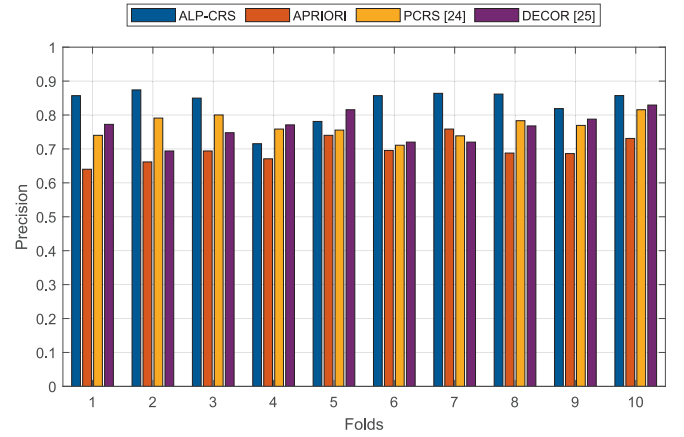


Fig. 4. Precision of Recommendation Systems Across Iterations (No. of Suggestions = 1).

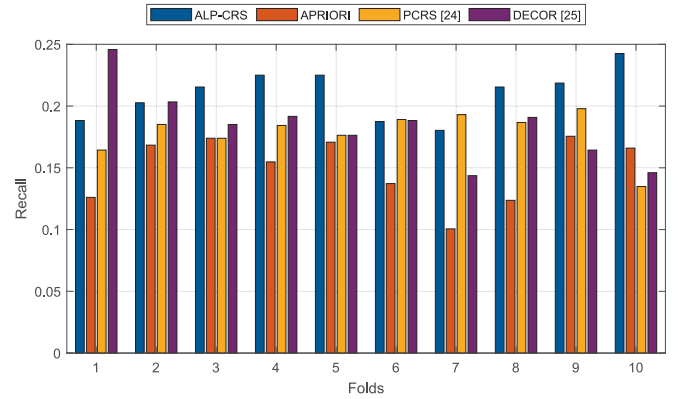


Fig. 5. Recall of Recommendation Systems Across Iterations (No. of Suggestions = 1).

capable of extracting students' actual choices than the APRIORI method, as seen by the improvement in recall. This improvement may be used as a measure to assess how effective relevant and helpful suggestions are in the real world. Recall curves also show the amount of suggestions based on a single recommendation, much as accuracy does.

Fig. 5 indicates that the normal values of recall in this experiment are generally less than 0.25. It is necessary to add that the values that were reported in this experiment were received when the number of recommendations was 1, whereas the average number of courses students chose is 3.96. Moreover, the suggested system employs strict confidence levels (0.65) and similarity filtering based on graphs to make sure that only the most useful courses are proposed. Although this method inherently restricts coverage (recall), it maximizes the trustworthiness of the provided advice. Thus, in cases where the recommender system only suggests a single suggestion, it is inherent that not all the options can be addressed. It will be demonstrated in the following experiment that the rate of recall will grow as more suggestions are given.

F-measure is shown in Fig. 6. The F-measure statistic, which combines the metrics of accuracy and recall, further suggests that the suggested strategy is more successful. This indicates that, in comparison to the approaches that were compared, the suggestions made by the suggested method are of a better caliber.

In Fig. 7, the precision metric is shown based on changes in the number of recommendations. In the previous phase, only one recommendation was taken into account. The indicators now indicate a diminishing trend as the number of referrals rises. This is because there is a greater chance of making suggestions that are unrelated when there are more recommendations. The effective recommendation system

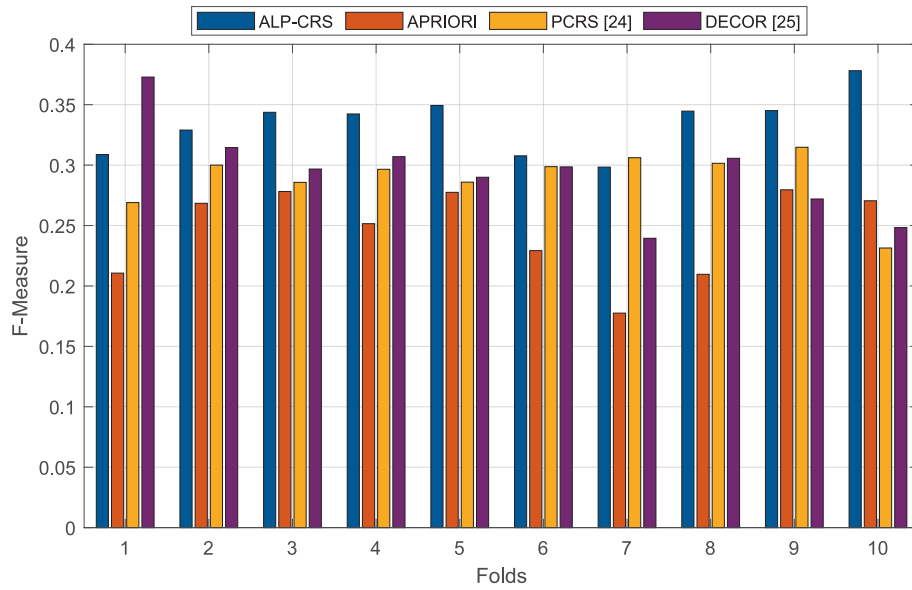


Fig. 6. F-measure of Recommendation Systems Across Iterations (No. of Suggestions = 1).

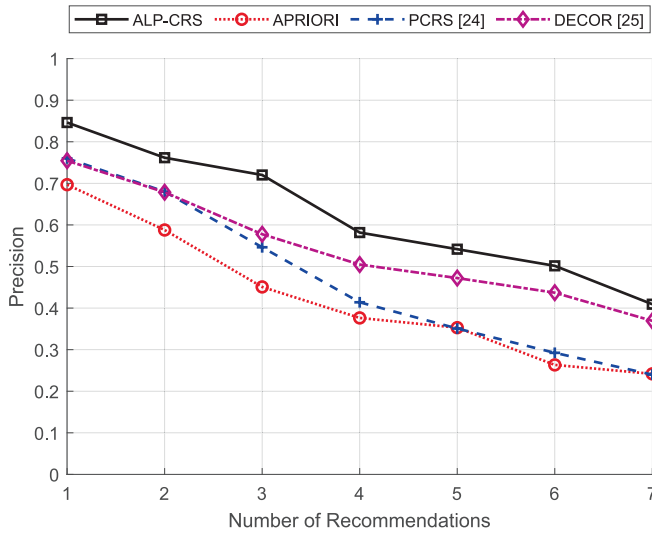


Fig. 7. Precision of Recommendation Systems for Varying Numbers of Recommendations (1–7).

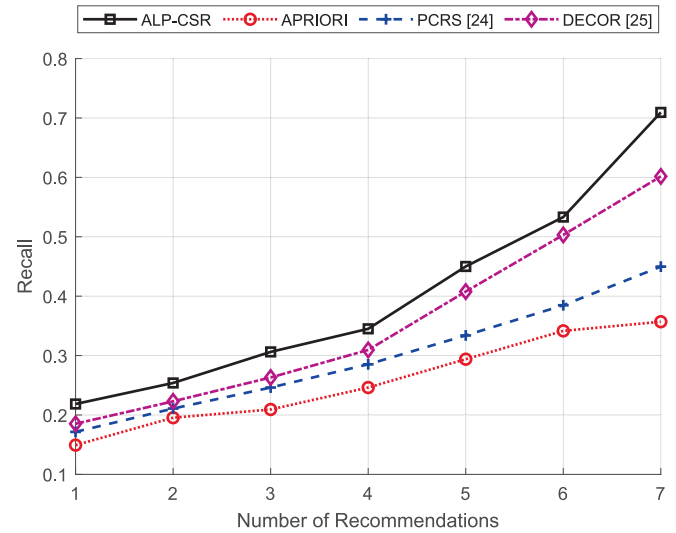


Fig. 8. Recall of Recommendation Systems for Varying Numbers of Recommendations (1–7).

ranks related items with a higher probability, resulting in the users preferred items being prioritized in the top recommendations. In comparison with other methods, the precision values are higher in various conditions, with a precision of approximately 0.4, which represents the highest precision.

The recall metric is shown in Fig. 8 as a function of variations in the number of suggestions. Regarding the recall measure, the tendency is in the other direction, indicating that as the number of recommendations grows, so does the recall value. The rationale for this is that the more suggestions we provide, the more likely it is that one of them will fall within the student's area of interest. This proves our previous claim about low recall values for one suggestion. By increasing the number of suggestions, the recommender system can cover a wider range of courses desired by the student; and this will lead to increasing recall rates. These results demonstrate that number of suggestions should be considered as a tunable parameter for providing a balance between precision and recall of the recommender system. Additionally, our suggested strategy has shown greater recall values under different settings, suggesting that

it takes into account a larger percentage of real user selections as suggestions. The results show that our suggested approach outperformed the APRIORI technique by 0.1% and the PCRS [24] method by 2.5%.

The F-measure metric is shown in Fig. 9 in relation to variations in the number of suggestions. The F-measure rises in tandem with the amount of suggestions. This suggests that the recommender system is becoming better; that is, the quality of the suggestions is getting better as the quantity of recommendations becomes higher. While the precision value decreases, attributed to the fixed nature of the set of domain choices, increasing the number of recommendations can lead to a simultaneous increase in domain-specific recommendations and an overall improvement in recommendation quality. Overall, the proposed method has shown better performance compared to the counterpart methods, APRIORI and DECOR [25], with values of 1.2% and 0.5%, respectively.

Although Precision, Recall, and F-measure are used to measure the overall accuracy of the recommendations, a complete recommender system should also be evaluated based on its capability to rank items

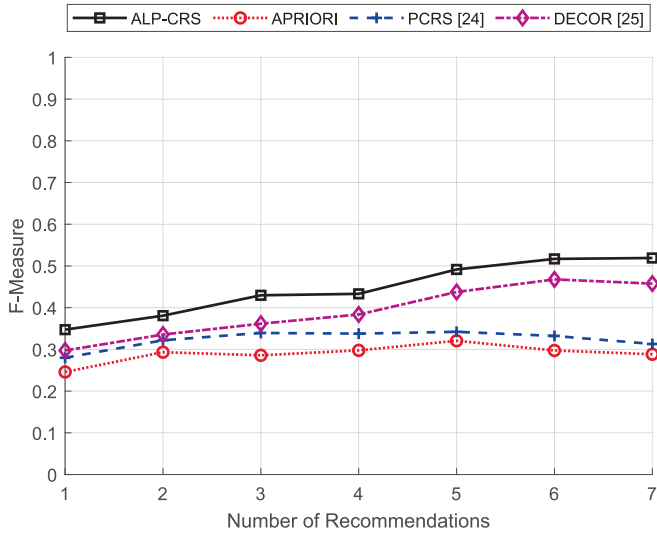


Fig. 9. F-measure of Recommendation Systems for Varying Numbers of Recommendations (1–7).

optimally and offer a wide range of recommendations. In order to measure these dimensions we used Mean Reciprocal Rank (MRR) and Catalog Coverage.

MRR is used to assess the distance the user has to scroll the recommendation list before finding a course that is relevant to them. It is determined by the following formula:

$$MRR = \frac{1}{|U|} \times \sum \frac{1}{rank_i} \quad (9)$$

where $|U|$ is the number of users, and $rank_i$ is the position of the first relevant recommended course to user i . The bigger the MRR, the more relevant courses are placed on the first position of the list by the system. Table 6 demonstrates that ALP-CRS has an MRR of 0.6807, which is better than the standard APRIORI method (0.5216) and the DECOR baseline (0.6111). This increase can be directly related to the final ranking score (Equation (5)) which is a fine-tuning of the global association rules based on local neighborhood similarities, so that the most contextually relevant courses are ranked highly.

Recommendation variety is a guarantee that the system does not just suggest the most popular courses globally to all students, which is a well-known drawback of standard association rule mining. Catalog Coverage is the percentage of all courses in the database that are suggested to at least one student during the entire period of evaluation:

$$Coverage = \frac{|I_{rec}|}{|I_{total}|} \times 100 \quad (10)$$

Where $|I_{rec}|$ is the distinct number of courses prescribed to all users and $|I_{total}|$ is the total number of courses available. In our experiments, standard APRIORI had a coverage of 34.17 because it is biased towards courses that have high global support thresholds. Conversely, the clustering stage of ALP-CRS enables the system to find niche, specialized courses which are pertinent to particular sub-communities. This enhanced the Catalog Coverage to 58.13 which showed that there was

great improvement in variety of recommendations without compromising on precision.

4.3. Sensitivity Analysis and Cold Start Evaluation

A sensitivity analysis was performed in order to fully assess the strength of the ALP-CRS model under various settings. The two operation cases that are specifically analyzed include the sensitivity of the algorithmic support threshold and the system performance when new users with no course history appear (the cold start problem).

5. A) Influence of Minimum Support Threshold (ϵ)

The APRIORI algorithm has a minimum support level (ϵ) that identifies the frequency at which a course selection pattern is regarded as a valid rule. We tested the system Precision, Recall and F-measure at three levels of support 0.1, 0.2 (baseline) and 0.3. Table 7 shows the outcomes of this analysis.

Table 7 indicates that reducing the threshold to 0.1 produces an enormous amount of rules, which raises recall (finding more potentially relevant courses) a little, but adds noise, reducing the precision dramatically. On the other hand, increasing the threshold to 0.3 will result in too strict rules; the precision will be high, but the recall will be lower since fewer valid recommendations will be produced. The optimal value of 0.2 gives the best balance, which validates the parameters used in the main experiments.

6. b) Cold Start Performance (New Users)

Probably one of the most important issues of recommendation architectures is how to make correct recommendations to new users who do not have a transactional history. ALP-CRS addresses this by creating a hypothetical connection between a new user and the closest existing student in terms of demographics, which essentially puts the new user in a pre-existing behavioral category (see Section 3.2.1).

To confirm this setting, an experiment was carried out that isolated a subgroup of New Users (simulated by not revealing their past course history) and compared their accuracy in making recommendations with Established Users (users with multiple terms of history). The selection of students in the new and the established users was random. Every category had 5 percent of the dataset individuals and the experiments were repeated 10 times. Table 8 shows the results of this experiment.

Table 8 results indicate the usefulness of the hypothetical demographic-based link. The system is able to use the behavioral trends of the given cluster to give very relevant recommendations even in the absence of longitudinal data. The precision rate of new users with 0.78 is within a 5 percent range of the established user precision of 0.82. This proves that ALP-CRS is highly reliable and effective irrespective of the length of tenure of the user in the educational system.

7. Discussion

This section presents a detailed evaluation of the results obtained in the experiments conducted, and it covers the effectiveness of the proposed method, its applications, and drawbacks. Further, we discuss the practical implications of our study and outline directions for future

Table 6
Detailed Assessment of Ranking and Variety (Top-7 Recommendations).

Method	Mean Reciprocal Rank (MRR)	Catalog Coverage (%)
APRIORI	0.5216	34.17
PCRS [24]	0.5834	49.56
DECOR [25]	0.6111	53.29
ALP-CRS (Proposed)	0.6807	58.13

Table 7
Minimum support (ϵ) effect on system performance.

Minimum Support (ϵ)	Precision	Recall	F-Measure	Observation
0.10	0.68	0.31	0.42	High noise, low relevance
0.20 (Baseline)	0.82	0.24	0.37	Optimal balance
0.30	0.85	0.15	0.25	Overly restrictive

Table 8

Performance comparison: existing users vs new users.

User Category	Precision	Recall	F-Measure
Established Users	0.82	0.24	0.37
New Users (Cold Start)	0.78	0.21	0.33

research to improve the effectiveness and usability of the course recommendation systems.

7.1. Performance Analysis

The results of the experiments show that the proposed method enables one to provide accurate recommendations to learners concerning the courses they should take online. It may be observed that the combination of link prediction and clustering methods has provided better precision and recall compared to the APRIORI baseline. Compared to association rules that only relate courses and students, the proposed method can discover more intricate patterns and preferences of students. When the model provides only one recommendation, the precision and recall of the proposed ALP-CRS is 0.84 and 0.24, respectively. This demonstrates 14.2% and 11.5% improvements compared to APRIORI in terms of precision and recall, respectively.

A closer examination of the trade-off between the high precision (82%) and the relatively lower recall (24%) will tell the mechanics behind the ALP-CRS architecture. In education, the recommendation of a course that is not appropriate (False Positive) may negatively impact the academic trajectory of a student; hence, the system is configured to be more specific. The low recall is largely due to two reasons: the student similarity graph has a low density and the confidence threshold (0.65) used in the rule mining stage is very high. The system reduces the search space to highly similar peers by filtering the candidates using graph clustering and link prediction (Phase 2). Although this rigor is what rules out some possible valid recommendations (reducing recall), it guarantees that the advice generated is very personalized and precise. This balance is in line with the strategies mentioned in [35], which state that in AI-driven distance learning, the evaluation metrics should be weighted based on the particular risks of the application domain (in this instance, the avoidance of misleading academic advice should be prioritized).

Moreover, these findings directly deal with the shortcomings of conventional recommender systems that were outlined in the introduction, including the absence of personalization in the general rule-based approaches. As demonstrated in [31], connecting the problem statement (specifically the inability of global rules to capture individual nuances) to the solution outcomes is vital for system validity. ALP-CRS addresses these drawbacks by incorporating the student similarity graph. Where a standard Apriori model would use a generic rule on all students, ALP-CRS narrows this down to the demographic and behavioral clusters that are created during Phase 2, which is helpful in reducing the one-size-fits-all error that is inherent in the traditional educational data mining.

Moreover, the comparison of the proposed method with the PCRS [24] and DECOR [25] methods demonstrate the effectiveness of the approach in question, in terms of precision, recall, and F-measure. The findings show that the proposed approach could improve f-measure by 5.5% and 4.7% compared to the PCRS [24] and DECOR [25] methods, respectively. Although, these methods utilize modern approaches, the proposed method here provides a simpler and more explainable approach with comparable performance. Additionally, the second experiment demonstrated that when the number of recommendations is increased to 7, the recall measure will reach 0.709. This proves the ability of our approach in retrieving desirable content during recommendation.

A statistical analysis was performed on the f-measure metric to show the significance of the differences between the proposed method and

compared approaches. One-way analysis of variable (ANOVA) was chosen for this purpose. The variations of f-measure for each method during iterations was structured as a vector and then the proposed approach was compared with each of the compared methods, one-by-one. The normality checks were conducted on each column of the generated matrix using the quantile-quantile (Q-Q) diagrams and the Shapiro-Wilk test. These tests' outcomes demonstrated that each model column's distribution is normal, allowing for the statistical examination of f-measure using one-way ANOVA. The Shapiro-Wilk test ($p > 0.05$ for the f-measure criteria) confirmed that the one-way ANOVA is appropriate for comparing the means of several groups when the data is regularly distributed.

The findings of the experiment indicate that the performance of the suggested strategy differs from each of the compared approaches in a statistically significant way ($p < 0.05$). The ANOVA test findings were interpreted as follows: ($F = 173.21, p = 4.26 \times 10^{-17}$) for the ALP-CRS compared to APRIORI, ($F = 86.52, p = 1.47 \times 10^{-13}$) for the proposed technique compared to PCRS [24], and ($F = 15.31, p = 1.14 \times 10^{-12}$) for the ALP-CRS compared to DECOR [25]. These findings also demonstrate the statistically significant discrepancies between the analyzed models and the suggested ALP-CRS, confirming that the advancements made in this method are not coincidental.

7.2. Practical Implications

The proposed course recommendation system has several implications for online education as follows:

- **Personalized Learning:** Such an approach allows for improving the quality of the learning process as the system takes into account personal preferences of every learner. It becomes possible to offer courses that a student may find interesting, courses that he or she already knows, or courses that may help the student achieve his or her learning objectives, thus increasing the students' interest and motivation.
- **Efficient Course Selection:** It will be easier for the students to find their way through the large number of courses since the system can recommend the most suitable ones. This may be helpful in that it eliminates the need for some work that a new student may not know how to do anyway.
- **Improved Student Success:** By suggesting courses that are appropriate, from a learner perspective, the system can enhance learner performance. Students will excel in the class if they complete course work that interests them and is easy for them to do.
- **Data-Driven Insights:** The system can provide important information about the students' behavior and preferences to support instructional and curriculum design, as well as resource management.

To address the scalability concern of the model, the proposed method can be scaled to handle larger datasets and increasing numbers of users by:

- **Distributed Computing:** The advantage being the use of distributed computing frameworks such as Apache Spark to bring computations and large data set processing into parallel.
- **Incremental Learning:** Applying methods to fit the model with new data gradually without requiring the use of entire data set again leading to a considerable saving in the processing time.
- **Offline Learning:** The proposed link prediction approach supports an offline phase for the similarity calculation phase. In this case, any increase in processing time is entirely abstracted away from the user. The heavy computations (APRIORI rule mining and community Link Prediction) are executed during the Offline Learning phase. The online stage only recalls these already computed matrices and makes sure that the system provides real-time suggestions without taking

into account the overall number of enrolled students or courses. Using this approach, updating of the model can be executed in the background without making any interrupt in the operations of the recommender system. Also, as discussed in [section 3](#), decomposing the graph structure into several clusters is very effective in reducing the computational complexity of the model.

- **Model Optimization:** Tuning the model design and the corresponding parameters in order to increase its speed and expandability. For example, reducing the maximum path length in the like prediction algorithm can increase the speed of model construction phase for large datasets.
- **Generalizability and Cross-Domain Applicability:** Even though the existing validation is based on a particular institutional dataset, the ALP-CRS framework is designed to have a domain-agnostic architecture. The underlying reasoning (learning global behavioral patterns using APRIORI and optimizing them using local graph-based similarities) does not depend on course titles or institutional organization. In order to implement the system in another educational setting, e.g., in a MOOC or in another university, all that is needed is the re-calibration of the student feature vector x in Equation (3). The given vector can be easily extended to incorporate various characteristics that are applicable to other cohorts, including professional experience, language proficiency, or different credit systems. Since the clustering step is dynamically adjusted to the underlying topology of any particular user graph, the model can by default detect similar neighbors irrespective of the discipline, which validates its universal applicability potential.

By so doing, the proposed system can be implemented in large scale online learning environments as hinted above.

8. Conclusion

This paper introduced ALP-CRS, a new course recommendation system that can improve personalization in online learning by combining association rule mining and graph analysis. The proposed method overcomes the weakness of the traditional static rule-based systems because it extracts behavioral patterns using the APRIORI algorithm and refines them using student similarity graphs. The empirical findings showed a precision rate of 82% and a recall rate of 24 which is much better than comparative methods such as PCRS and DECOR.

Such results can be extended to the implementation of data-driven educational technologies. The precision is high, so the learners do not have to be overwhelmed with irrelevant options, which addresses the problem of information overload in online platforms directly. Moreover, the fact that the system uses graph clustering to break down the student population into local communities provides a viable solution to scalability. This architectural design enables the system to handle large scale datasets effectively and therefore it is applicable when deploying the system in large online learning environments where computational efficiency is the most critical factor.

8.1. Limitations and Future Directions

The following section presents the study's limitations and discuss the possible avenue for future research. Despite the fact that the proposed method has given positive results, there are some drawbacks as well:

- **Data Limitations:** This study used a single dataset from one university for the experiments. This limitation may restrict the generalizability of the findings to other educational institutions with diverse student populations and course offerings.
- **Cold Start Problem:** Although the proposed method offers a graph-based solution for facing the cold start problem, but this approach needs a high number of dataset instances to solve the mentioned issue. The system may have a problem of recommending items that

the new user may not be interested in due to low number of interactions with the system.

- **Scalability:** While the number of users as well as the number of courses is increasing, the complexity of the graph-based algorithms might become a problem. Although combining link prediction with the clustering approach mitigates this issue to some extent, it remains a potential limitation for large-scale applications.

Directions for future research are as follows:

- **Incorporating Contextual Factors:** Including more contextual factors while forming the graph model of the recommender system could improve its flexibility. Therefore, exploring the impact of contextual factors could be a topic for future studies.
- **Hybrid Approaches:** Integration of graph-based methods with other types of recommender methods including user-user collaborative filtering and item-item collaborative filtering and content-based filtering could be investigated in future works.
- **Active Learning:** Exploring solutions for receiving user feedback in order to improve the recommendations could be studies in future works.
- **Generalizability:** Evaluating the model on datasets which cover a wider range of courses and include a larger set of instances could be beneficial for problems such as generalizability.

Data availability

All data generated or analysed during this study are included in this published article.

Ethical Statement

"This study did not involve human subjects research as defined by Belmont Report. Data was collected through a voluntary anonymous questionnaire that did not pose any risks beyond those encountered in daily life. Informed consent was obtained from all participants."

CRediT authorship contribution statement

YanJun Gao: Investigation. **Xiaoyang Liu:** Investigation. **Xinke Gao:** Investigation. **Junming Zhang:** Investigation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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